Consistency

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- We say T_n is a **consistent** estimator of θ if

$$T_n \xrightarrow{p} \theta.$$

- $\bar{X}_n$ is a consistent estimator of μ.
- The sample variance S^2 is a consistent estimator of σ^2.

Consistency vs. Unbiasedness

- Consistency is a very important property from an estimator to have. It is a poor estimator that does not approach its target as the sample size gets large. Note that the same cannot be said for the property of unbiasedness. For example, instead of using S^2 to estimate σ^2, we use $V = n^{-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$. Then V is consistent for σ^2, but it is biased because $E(V) = (n-1)\sigma^2/n$. The bias of V is σ^2/n which vanishes as $n \to \infty$.

How to show consistency?

- WLLN
- Definition
- Chebyshev's Inequality

For a continuous random variable T_n,

$$\mathbb{P}(|T_n - \theta| < \epsilon) = \mathbb{P}(\theta - \epsilon < T_n < \theta + \epsilon) = F_{T_n}(\theta + \epsilon) - F_{T_n}(\theta - \epsilon)$$

Examples

- Let the random variable Y_n have a distribution that is $b(n, p)$. Prove that Y_n/n converges in probability to p. This result is one form of the weak law of large numbers.

- **Maximum of a Sample from a Uniform Distribution**: Let $X_1, \ldots, X_n$ is a random sample from Uniform$(0, \theta)$ distribution. Let $Y_n = \max(X_1, \ldots, X_n)$. Show that $Y_n \xrightarrow{p} \theta$. Note that unbiased estimator $((n+1)/n)Y_n$ is also consistent.

Examples

Examples

- Let $X_1, \ldots, X_n$ be iid from $N(\mu, 1)$.
  — What is an unbiased estimator for μ^2?
  — What is a consistent estimator for μ^2?

Examples

References

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[3] *A First Course in Probability* by Ross

[4] *John E. Freund's Mathematical Statistics with Applications* by Irwin Miller Marylees Miller